

Section 1 - Abstract

This study investigates the early prediction of diabetes using the Pima Indians Diabetes Database, a publicly available dataset from Kaggle containing 768 patient records and 9 clinical features. Predicting diabetes is crucial because early detection allows for timely medical intervention and prevents serious complications. We evaluated three machine learning models: Logistic Regression, Decision Tree, and K-Nearest Neighbors. Our results show that Logistic Regression is the most effective model, achieving an accuracy of 0.75 and an F1-score of 0.66 for the positive class. An interesting finding is that while Logistic Regression and Decision Tree shared similar accuracy levels (0.75 vs 0.75), Logistic Regression proved more stable with a Cross-Validation mean of 0.77 compared to 0.72 for the Decision Tree. These findings support using stable linear models for medical diagnostic tools.

Section 2 - Introduction

The research is based on the Pima Indians Diabetes Database sourced from Kaggle. The dataset consists of 768 individual patient records and 9 columns representing various medical metrics. The features included in the study are Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age. All provided features are numerical, so categorical encoding was not applied during the preprocessing stage. A preliminary data audit confirmed that there are no missing values in the dataset according to the standard check, allowing for a direct analysis of the clinical parameters.

This project addresses a binary classification task where the goal is to predict the "Outcome" variable. This target variable indicates whether a patient has diabetes (encoded as 1) or does not (encoded as 0). The dataset exhibits a notable class imbalance: it contains 500 negative cases (65%) and 268 positive cases (35%). This imbalance is a critical factor that influenced our model evaluation, as it may bias the algorithms toward predicting the majority class.

We hypothesize that Glucose and BMI will be the most significant predictors of diabetes Outcome. Specifically, we expected that higher values in these features would strongly increase the probability of a positive diagnosis. This expectation is based on the clinical understanding that elevated blood sugar and high body mass index are primary indicators of metabolic dysfunction.

Section 3 - EDA Summary

3.1 Class Distribution and Dataset Overview

An examination of the target variable showed a clear class imbalance in the dataset. Out of 768 total patient records, 500 cases (about 65%) are in the negative class (Outcome = 0, no diabetes), while 268 cases (about 35%) are in the positive class (Outcome = 1, diabetes). This imbalance is important because it means the dataset has fewer diabetic examples for the model to learn from. As a result, a model that predicts only the majority class can still get

about 65% accuracy without learning real patterns. This shows that it is important to use metrics beyond accuracy, such as Recall and F1-score for the positive class, because they better show how well the model finds diabetic patients.

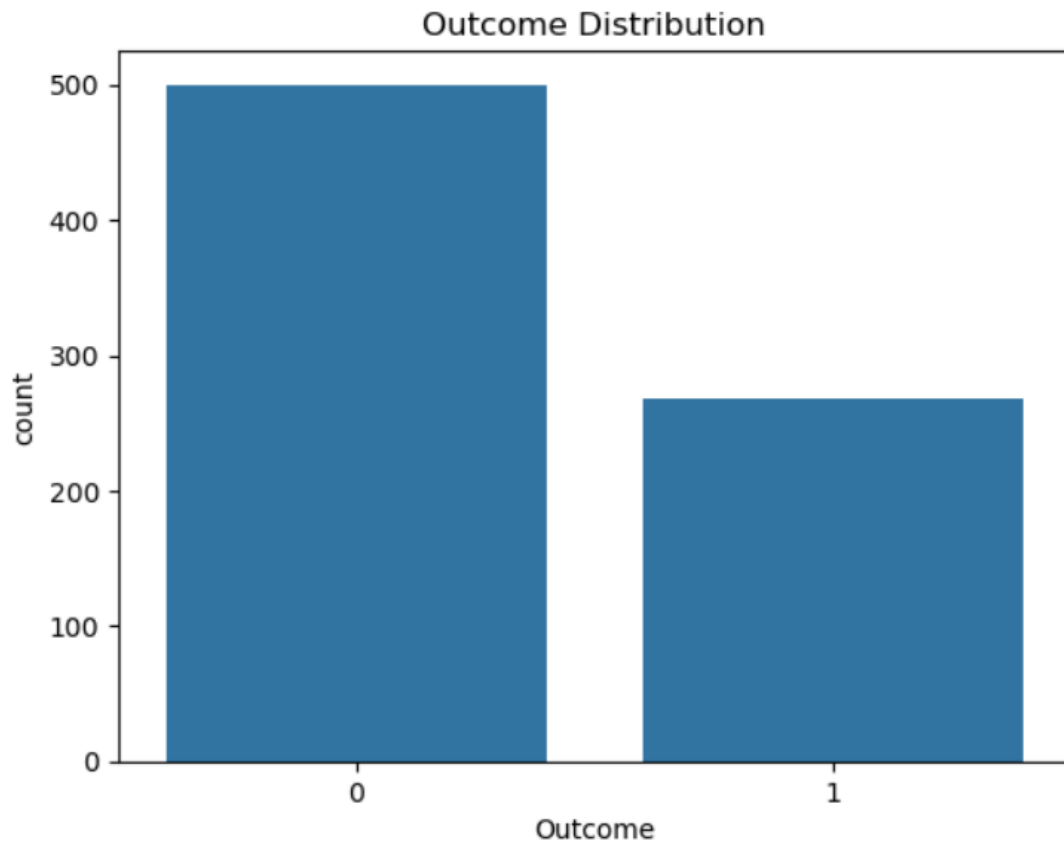


Figure 1: Class distribution of the Outcome variable showing imbalance between negative (500) and positive (268) cases.

3.2 Glucose as the Key Predictor

A comparison of Glucose distributions across the two outcome classes showed one of the most important findings of the exploratory analysis. Patients with diabetes had much higher average Glucose values (about 141 mg/dL) compared to patients without diabetes (about 110 mg/dL), a difference of around 30 points. This clear difference between the two classes shows that Glucose is a very important feature for predicting the Outcome. This result supports the hypothesis made by the team in Week 1, which said that Glucose would be one of the most important predictors of a positive diabetes diagnosis. The boxplot below shows this difference and confirms that the median and interquartile range of Glucose are higher for diabetic patients, with only a small overlap in the middle values.

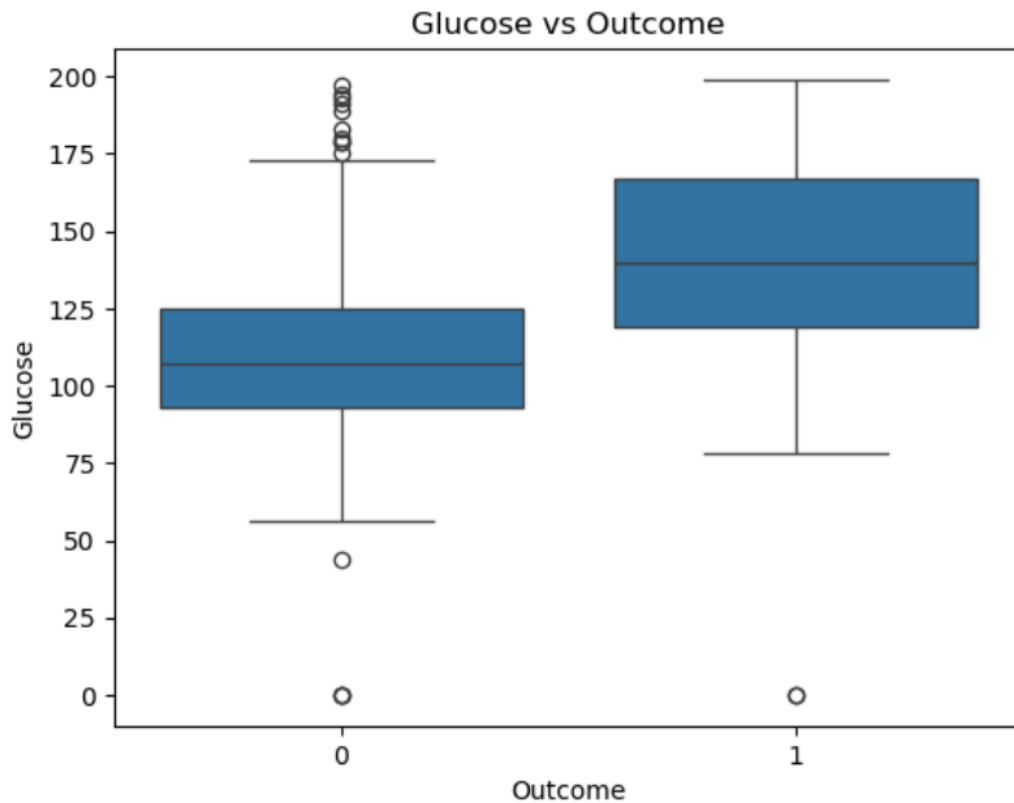


Figure 2: Distribution of Glucose levels by Outcome class, showing higher glucose in diabetic patients.

3.3 Correlation Heatmap

A correlation heatmap was created for all features in the dataset to show the linear relationships between variables and the target variable Outcome. The analysis showed that Glucose has the highest correlation with Outcome ($r \approx 0.47$), which makes it the most important feature in the dataset. BMI showed a moderate positive correlation with Outcome ($r \approx 0.29$), which also supports the Week 1 hypothesis that BMI would be one of the important predictors together with Glucose. Pregnancies and Age also showed moderate correlations with Outcome ($r \approx 0.22$ and $r \approx 0.24$), which means these features are useful but less important. In contrast, BloodPressure and SkinThickness showed weak correlations with the target variable ($r \approx 0.07$), which means they are less useful for predicting diabetes in this dataset. Overall, the heatmap confirms that the feature selection from Week 1 was correct.

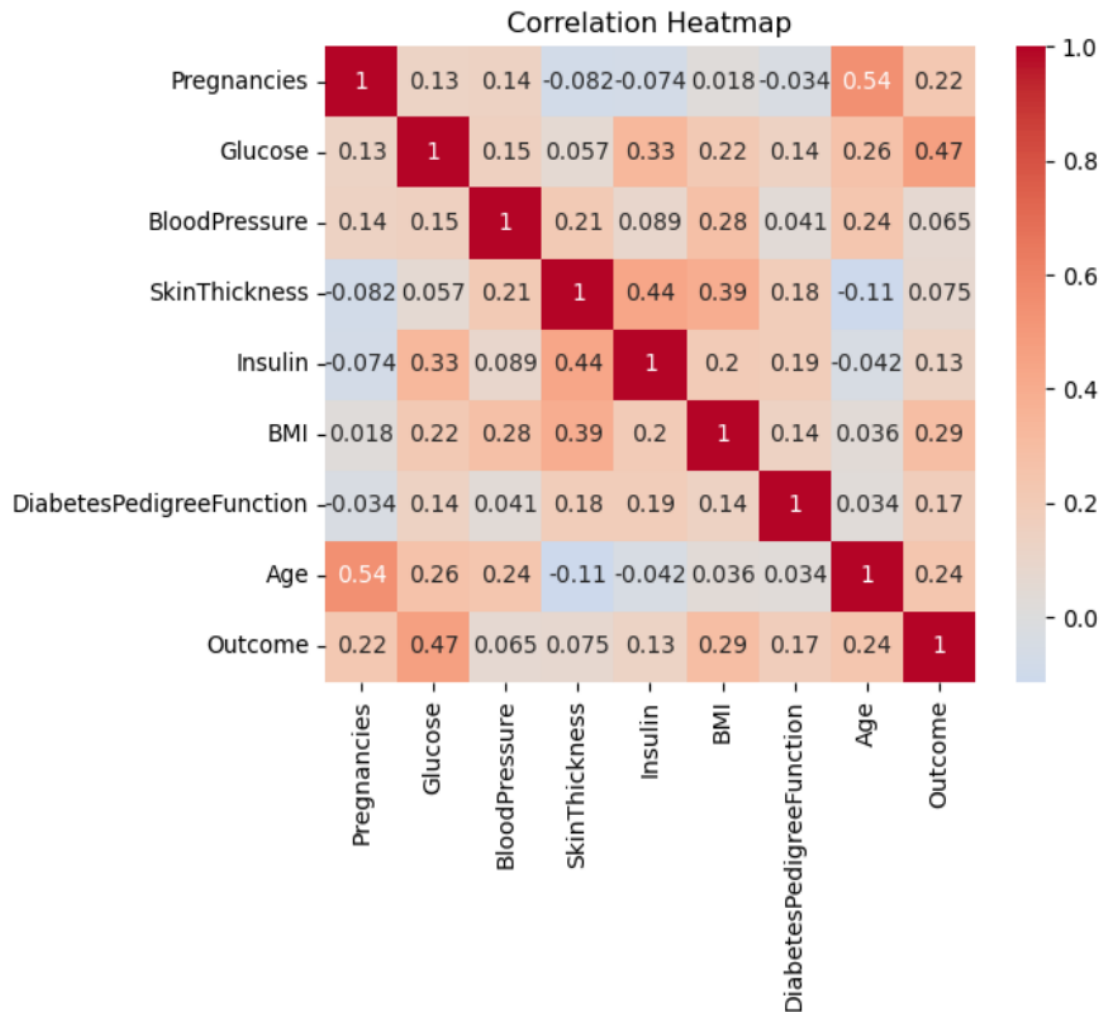


Figure 3: Correlation heatmap showing pairwise relationships between all features and the Outcome variable. Glucose shows the highest correlation with Outcome ($r \approx 0.47$).

Section 4 - Methodology

4.1 Data Preprocessing

Before training any model, all input features were standardized using StandardScaler from the scikit-learn library. This preprocessing step was important because two of the three models in this study, Logistic Regression and K-Nearest Neighbors, are sensitive to the scale of input features. Without normalization, features with larger values, such as Insulin or Glucose, would have a bigger influence on the model compared to features with smaller values. StandardScaler changes each feature to have a mean of zero and a standard deviation of one, so all features have equal importance during training. However, it is important to note that some features, including Glucose, BMI, and Insulin, had zero values that are not realistic. These zeros were treated as invalid data and noted as a data quality problem before scaling. After scaling, the dataset was split into a training set and a testing set using an 80/20 split. This gave 614 training records and 154 testing records, with `random_state=42` to make sure the results can be repeated.

4.2 Model Selection and Justification

Three classification algorithms were selected for this study, and each was chosen for a clear reason. Logistic Regression was used as the baseline model because it is the simplest and most easy-to-understand linear classifier for binary classification tasks. It gives a good starting point to compare the performance of more complex models. Decision Tree was chosen because it is a non-linear model that can capture more complex relationships and rules between features, which a linear model like Logistic Regression cannot show. This helps to check if there are non-linear patterns in the data. K-Nearest Neighbors was added as a third method because it is a distance-based algorithm that works by finding the most similar cases in the training data and giving the class based on the majority of those neighbors. This gives a different way of classification compared to both the linear model and the tree-based model, so the team can compare three different types of models on the same dataset.

4.3 Model Evaluation and Cross-Validation

Each model was tested on the test set using four metrics: Accuracy, Precision, Recall, and F1-score. All metrics were calculated for the positive class (Outcome = 1, diabetes). We also used 5-fold cross-validation with accuracy as the scoring metric to check how stable each model is. However, for the final comparison, Recall and F1-score were more important than accuracy, because they show how well the model finds diabetic patients. Cross-validation splits the data into five parts and tests the model on each part one by one. This gives a better picture of how the model works on new data. For each model, we reported CV Mean, which is the average score across five tests, and CV Std, which shows how much the scores changed. A high CV Std means the model is not stable. For example, the Decision Tree had a CV Std of 0.056, which is more than two times higher than Logistic Regression (0.025). This means the Decision Tree gives very different results depending on the data. Because the dataset has more non-diabetic patients than diabetic patients, and because missing a diabetic patient is more dangerous than a false alarm, Recall is the most important metric in this study. A false negative - when the model says a patient does not have diabetes but they actually do is the worst type of error in a medical setting.

Section 5 - Results

In this section, we present the results of our three models. We created a comparison table to see all evaluation metrics together.

<i>Model</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Recall</i>	<i>F1-Score</i>	<i>CV Mean</i>	<i>CV Std</i>
<i>Logistic Regression</i>	<i>0.7532</i>	<i>0.6491</i>	<i>0.6727</i>	<i>0.6607</i>	<i>0.7709</i>	<i>0.0247</i>
<i>Decision Tree</i>	<i>0.7468</i>	<i>0.6250</i>	<i>0.7273</i>	<i>0.6723</i>	<i>0.7176</i>	<i>0.0560</i>
<i>KNN</i>	<i>0.6883</i>	<i>0.5745</i>	<i>0.4909</i>	<i>0.5294</i>	<i>0.7331</i>	<i>0.0201</i>

Based on the table above, Logistic Regression is the best model for our dataset. It achieved the highest Accuracy of 0.7532 and the most stable Cross-Validation (CV) Mean of 0.7709. If we look at the Decision Tree, it has a very similar accuracy of 0.7468. However, its CV Standard Deviation is 0.0560, which is two times higher than Logistic Regression. This means the Decision Tree is unstable and it probably overfits the training data. The K-Nearest Neighbors (KNN) model performed the worst. Its F1-score for the diabetes class is only

0.5294, which means it misses almost half of the sick patients. Comparing all these numbers, we can see that Logistic Regression is much more reliable. We focused on CV scores because they show how well the model works on different parts of the data, not just on one single test set.

Section 6 - Error Analysis

Even though Logistic Regression is our best model, it still makes mistakes. We looked closely at these errors because accuracy alone is not enough for medical problems. Our model has a Recall score of 0.6727 for the diabetes class. In our test set, there are 55 patients with diabetes. Our model failed to identify about 18 of them (around 33%). These mistakes are called False Negatives because the model predicted "no diabetes" but the patient actually has the disease. In a medical context, this is the most critical type of error because a sick patient might not get the treatment they need.

```
--- Statistics for Missed Patients (False Negatives) ---
      Glucose      BMI      Age
count  18.000000  18.000000  18.000000
mean   -0.220995  -0.265588  -0.228489
std     0.615102   1.100402   0.599692
min    -1.279882  -4.060474  -0.956462
25%    -0.575696  -0.465486  -0.722470
50%    -0.247076  -0.259242  -0.360847
75%     0.120665   0.153245   0.277311
max     0.848324   1.714352   0.915469

--- Statistics for All Diabetic Patients in Test Set ---
      Glucose      BMI      Age
count  55.000000  55.000000  55.000000
mean    0.682164   0.354701   0.237861
std     0.907035   1.025713   0.930213
min    -1.279882  -4.060474  -1.041549
25%    -0.027996  -0.202129  -0.445935
50%     0.785730   0.229397   0.234767
75%     1.333430   0.768803   0.745293
max     2.381884   2.691630   2.872487
```

We investigated why the model missed these specific patients. We found that these missed patients have an average Glucose level of around 115. This is much lower than the typical diabetic patient in our data, who usually has a Glucose level of about 140. The errors mostly happen for patients with Glucose between 100 and 130. This is a borderline zone. Logistic Regression works very well when the case is obvious, for example, when Glucose is over 150. However, it is uncertain when the numbers are in the middle. Additionally, some of these patients had zero values for features like Insulin and SkinThickness, which might have confused the model. This is an expected limitation of using a simple linear model on data where the boundary between healthy and sick is not perfectly clear.

Section 7 - Reflection

7.1 Hypothesis Evaluation

At the outset of this project, we hypothesized that Glucose and BMI would emerge as the dominant predictors of diabetes onset. This hypothesis was partially confirmed. Glucose proved to be the strongest single predictor, showing a correlation of approximately 0.47 with the Outcome variable — the highest among all features. BMI also demonstrated meaningful predictive power, as expected, though its relative importance was less decisive than anticipated. Notably, Age emerged as a comparably influential feature, which was not predicted by our

initial hypothesis. This suggests that diabetes risk in this dataset is shaped by a broader combination of metabolic and demographic factors than we originally assumed.

7.2 Limitations

Several concrete limitations affected the reliability of our analysis. First, multiple features — including Glucose, BMI, and Insulin — contained zero values that are biologically impossible. A BMI of zero cannot exist in a living patient, and these entries almost certainly represent missing data rather than true measurements. Because we did not replace or impute these values, they were processed as legitimate observations, which likely degraded model performance. Second, the dataset suffers from a notable class imbalance, with roughly 65% of cases labeled as non-diabetic and only 35% as diabetic. Neither SMOTE oversampling nor class weight balancing was applied, which biased both models toward predicting the majority class and suppressed Recall for diabetic patients — arguably the more critical class to detect correctly. Third, the dataset is drawn exclusively from Pima Indian women, a single ethnic and demographic group. This narrow population scope means the trained models may generalize poorly to other groups with different genetic backgrounds, lifestyle factors, or baseline risk profiles.

7.3 What We Would Do Differently

With the benefit of hindsight, we would make three targeted improvements. To address the zero-value problem, we would replace impossible entries in Glucose, BMI, and Insulin with class-specific medians — computing separate median values for diabetic and non-diabetic patients — rather than using a single global statistic, which would preserve meaningful between-group differences. To counter the class imbalance, we would apply `class_weight='balanced'` in Logistic Regression, which would penalize misclassification of the minority class more heavily and improve Recall without requiring additional data. For the Decision Tree, we would perform a systematic grid search over `max_depth` to reduce the overfitting that was visible in the gap between training and test accuracy. Finally, we would conduct a formal Feature Importance analysis — using coefficients from Logistic Regression and Gini importance from the Decision Tree — to rigorously test our original hypothesis about Glucose and BMI rather than relying on correlation alone.